

Electrical Panels: Process, Flow & Optimization Overview

Purpose:

Electrical panels (distribution boards) serve as the central hub for controlling and distributing electricity within facilities. They house breakers, disconnects, and protective devices, ensuring safety, load balance, and system integrity.

System Start-to-Finish Flow:

1. ****Incoming Power Supply****
 - Power enters from utility mains or transformers at the service disconnect.
 - Surge protection devices may be installed here.
2. ****Main Distribution Panel (MDP)****
 - Main breaker routes power to multiple branch circuits.
 - Load calculations ensure total amperage stays within panel rating.
3. ****Branch Circuits****
 - Each circuit breaker controls a specific load (lighting, HVAC, pumps, outlets).
 - Circuit wires leave panel to connected equipment.
4. ****Safety Controls & Monitoring****
 - GFCI, AFCI, and lockout-tagout systems ensure user protection.
 - Periodic thermal imaging or load logging is used for diagnostics.

Key Components & Definitions:

- ****Main Breaker****: Disconnects entire panel from incoming power.
- ****Busbar****: Conductive bars that distribute power within panel.
- ****GFCI/AFCI****: Protection against shocks (GFCI) or arcs (AFCI).
- ****Lockout/Tagout****: OSHA-required method for isolating circuits during maintenance.
- ****Load Balancing****: Distributing electrical loads evenly across phases.

Example Flow:

> A 480V three-phase panel distributes power to 12 circuits. Circuit #4 feeds a 10 HP pump via a 30A breaker and disconnect. Thermal scans during monthly inspection reveal a hot spot near the neutral bus, prompting torque check and breaker replacement.